A CRITERION FOR THE NON-EMPTINESS OF THE REGULAR PART OF A QUIVER VARIETY IN TERMS OF ROOTS AND WEIGHTS

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Allowed standard prerequisites (editorial).

- The moduli space M_zeta(v,w) of stable modules, its regular part, and the stability parameter zeta with the Cartan form pairing p(x) = 1 - (1/2) x-transpose C x. Set up in the author's Sections 1 and 2, before the counted statement. Fixes every object of the two numbered parts, including the quantity p and the roots appearing in the inequalities.
- The classification of simple modules via the cited theorem of CB, and the hyper-Kahler moment map description of stable modules. Used inside the counted proof at source lines 1254-1259, where the author invokes Theorem 1.2 of the cited paper CB. Supplies the criterion for the existence of a simple

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module with prescribed moment map, which is the whole content of part (1).
- The construction of the extended quiver and the root-space identification for the extended graph. The author's lemma labelled lem:root (source lines 1184-1196), which the counted proof invokes. Provides the dimension-vector and root bookkeeping used in part (2).

Source reference. Original lemma 2.10: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1184–1184. Navigation pointer only: the counted proof invokes the author's root-space lemma in its part (2); that lemma has its own card in this lane and its text is not reproduced here.

Theorem 0.1. (1) *Suppose* $W = 0$ *and consider a dimension vector* $\mathbf{v}$ *with* $\zeta \cdot \mathbf{v} = 0$. *Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta}(\mathbf{v}, 0) \neq \varnothing$ *if and only if the following holds*:

- $\mathbf{v}$ *is a positive root, and* $p(\mathbf{v}) > \sum_{t=1}^{r} p(\beta^{(t)})$ *for any decomposition* $\mathbf{v} = \sum_{t=1}^{r} \beta^{(t)}$ *with* $r \geq 2$ *and* $\beta^{(t)}$ *a positive root with* $\zeta \cdot \beta^{(t)} = 0$ *for all* t,

where $p(x) = 1 - \frac{1}{2}{}^t x \mathbf{C} x$.

(2) *Suppose* $\mathbf{w} \neq 0$. *Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta}(\mathbf{v}, \mathbf{w}) \neq \varnothing$ *if and only if the following holds:*

- $\mathbf{w} - \mathbf{v}$ *is a weight of the integrable highest weight representation* $V(\mathbf{w})$ *of the highest weight* $\mathbf{w}$, *and* ${}^t\mathbf{v}(\mathbf{w} - \frac{1}{2}\mathbf{C}\mathbf{v}) > {}^t\mathbf{v}^0(\mathbf{w} - \frac{1}{2}\mathbf{C}\mathbf{v}^0) + \sum_{t=1}^{r} p(\beta^{(t)})$ *for any decomposition* $\mathbf{v} = \mathbf{v}^0 + \sum_{t=1}^{r} \beta^{(t)}$ *with* $r \geq 1$, $\mathbf{w} - \mathbf{v}^0$ *is a weight of* $V(\mathbf{w})$, *and* $\beta^{(t)}$ *a positive root with* $\zeta \cdot \beta^{(t)} = 0$ *for all* t.

Proof. (1) A ζ-stable module with $\mu = 0$ corresponds to a point satisfying the hyper-Kähler moment map $(\mu_{\mathbb{R}}, \mu) = (\zeta, 0)$, which correspond a 0-stable module (i.e., a simple module) with $\mu = \zeta$ after a rotation of the complex structures. Then by [8, Theorem 1.2] such a simple module exists if and only if the above condition holds.

(2) For the given $\mathbf{w}$ we construct the quiver $(\tilde{I}, \tilde{E})$ as above. We consider the dimension vector $\tilde{\mathbf{v}} = \mathbf{v} + \alpha_\infty$ and define the stability parameter $\tilde{\zeta} = (\zeta, \zeta_\infty)$ by $\zeta \cdot \dim \mathbf{v} + \zeta_\infty = 0$. Then $\mathfrak{M}^{\mathrm{s}}_{\zeta}(\mathbf{v}, \mathbf{w})$ is isomorphic to the quiver variety $\mathfrak{M}^{\mathrm{s}}_{\tilde{\zeta}}(\tilde{\mathbf{v}}, 0)$ associated with $(\tilde{I}, \tilde{E})$ [8, the end of Introduction]. Therefore we can apply the criterion in (1) together with the observation Lemma 2.10. We first note that

$$p(\tilde{\mathbf{v}}) = 1 - \tfrac{1}{2}{}^t\tilde{\mathbf{v}}\tilde{\mathbf{C}}\tilde{\mathbf{v}} = {}^t\mathbf{v}\left(\mathbf{w} - \tfrac{1}{2}\mathbf{C}\mathbf{v}\right).$$

Also if we have a decomposition $\tilde{\mathbf{v}} = \sum \beta^{(t)}$, one of $\beta^{(t)}$ has 1 in the entry ∞, and other $\beta^{(t)}$'s have 0 in the entry ∞. We rewrite the former as $\mathbf{v}^0 + \alpha_\infty$, and identify the latter with positive roots for (I, E). Now the assertion follows from (1). □

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